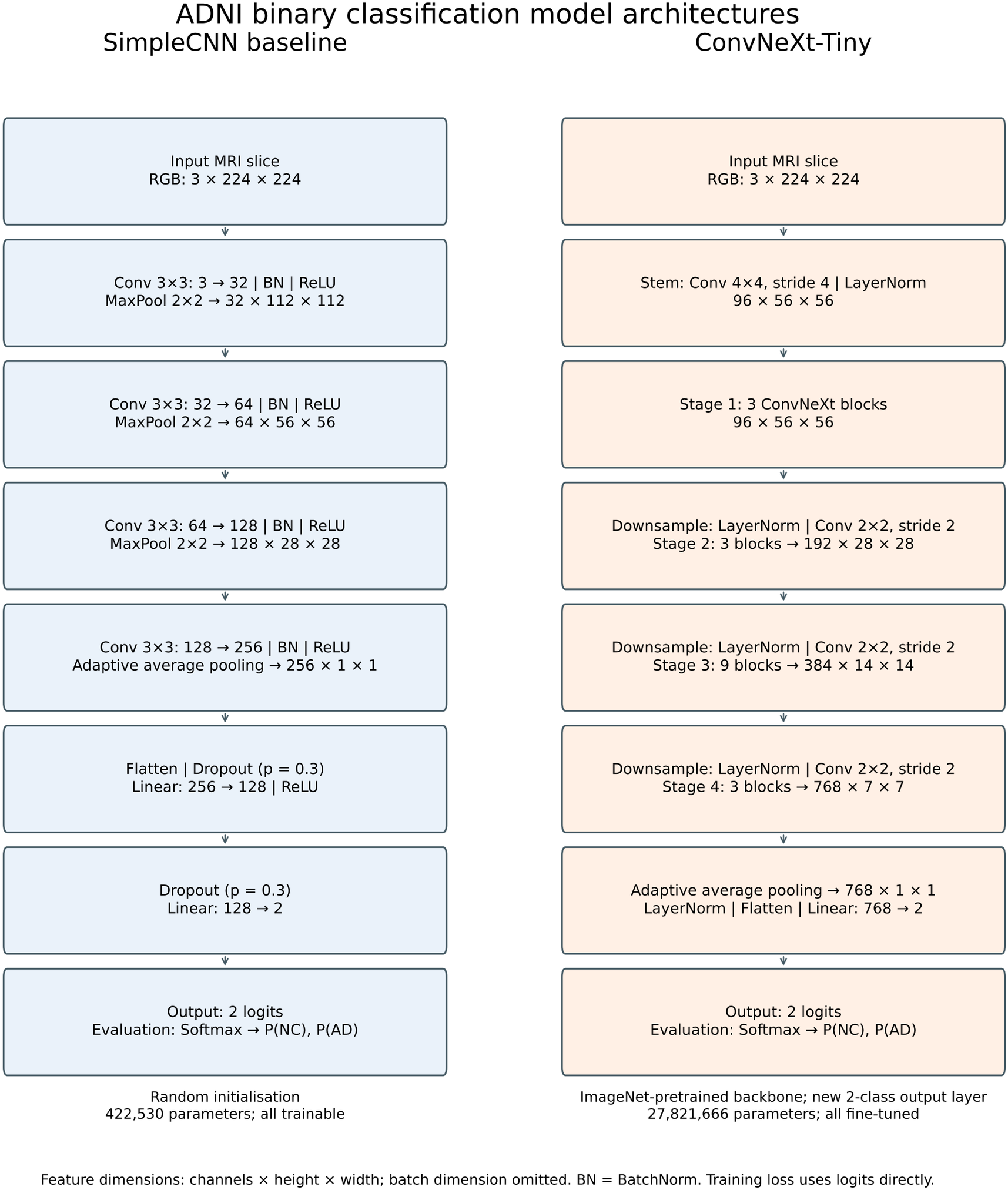
ADNI binary classification model architectures
SimpleCNN baseline
Input MRI slice
RGB: 3 × 224 × 224
Conv 3×3: 3 → 32 | BN | ReLU
MaxPool 2×2 → 32 × 112 × 112
Conv 3×3: 32 → 64 | BN | ReLU
MaxPool 2×2 → 64 × 56 × 56
Conv 3×3: 64 → 128 | BN | ReLU
MaxPool 2×2 → 128 × 28 × 28
Conv 3×3: 128 → 256 | BN | ReLU
Adaptive average pooling → 256 × 1 × 1
Flatten | Dropout (p = 0.3)
Linear: 256 → 128 | ReLU
Dropout (p = 0.3)
Linear: 128 → 2
Output: 2 logits
Evaluation: Softmax → P(NC), P(AD)
Random initialisation
422,530 parameters; all trainable
ConvNeXt-Tiny
Input MRI slice
RGB: 3 × 224 × 224
Stem: Conv 4×4, stride 4 | LayerNorm
96 × 56 × 56
Stage 1: 3 ConvNeXt blocks
96 × 56 × 56
Downsample: LayerNorm | Conv 2×2, stride 2
Stage 2: 3 blocks → 192 × 28 × 28
Downsample: LayerNorm | Conv 2×2, stride 2
Stage 3: 9 blocks → 384 × 14 × 14
Downsample: LayerNorm | Conv 2×2, stride 2
Stage 4: 3 blocks → 768 × 7 × 7
Adaptive average pooling → 768 × 1 × 1
LayerNorm | Flatten | Linear: 768 → 2
Output: 2 logits
Evaluation: Softmax → P(NC), P(AD)
ImageNet-pretrained backbone; new 2-class output layer
27,821,666 parameters; all fine-tuned
Feature dimensions: channels × height × width; batch dimension omitted. BN = BatchNorm. Training loss uses logits directly.